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CSIT495 Senior Project in Information Technology

Mwasalati

An Intelligent Online Public Bus Transportation

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Abstract

The lack of organized and accessible information about public bus routes in Yemen presents a significant challenge for individuals who depend on public transportation for their daily mobility. In Sana'a, passengers often rely on personal experience or verbal guidance from drivers and passersby to determine which bus to take and how to reach their destinations. This informal approach can lead to confusion, delays, unnecessary travel costs, and reluctance to use public transportation, particularly for visitors and individuals unfamiliar with the city.

To address this problem, this project presents **Mwasalati: An Intelligent Online Public Bus Transportation Guidance System in Yemen**, a mobile application designed to help users navigate public bus routes in Sana'a efficiently and confidently. The application displays the user's current location, identifies nearby bus routes, and allows users to search for destinations and landmarks. It then determines the most suitable route, including direct trips and journeys that require transfers, and presents boarding points, drop-off points, and route visualization on an interactive map.

The project follows Agile Modeling (AM) as the system analysis and design methodology, while Scrumban is used to organize the development workflow. The mobile application is developed using Flutter and Dart, Firebase Firestore for cloud-based transportation data storage, and Google Maps API for geographic visualization and route rendering. A separate web-based administrative dashboard is included in the target system design to support the management of transportation data. Its implementation is planned for the next academic semester and is not part of the currently implemented mobile application.

The proposed system targets residents of Sana'a as well as visitors and individuals who are unfamiliar with the city's transportation network. By transforming informal transportation knowledge into a structured digital platform, Mwasalati reduces the time and effort required to plan trips, simplifies the use of public buses, and contributes to the digital transformation of local transportation services in Yemen.

The project demonstrates how mobile and geographic information technologies can be used to solve a real-world urban transportation problem and establishes a scalable foundation for future expansion to other cities and additional smart transportation features.

Chapter 1

Introduction

1.1 Project Background

Public transportation plays a vital role in the daily lives of people in large cities, where many residents depend on buses to reach universities, workplaces, hospitals, and commercial areas. In Sana'a, public buses are among the most affordable and widely used means of transportation. However, navigating the bus network remains challenging because there is no dedicated digital system that provides clear and organized information about local routes.

Currently, passengers rely on personal experience or ask drivers and passersby to determine which bus to take, where to board, and where to get off. This approach is often inconvenient and unreliable, particularly for visitors and individuals who are unfamiliar with the city. As a result, users may spend significant time searching for suitable transportation options, choose inefficient routes, and incur unnecessary travel costs.

Recent advances in mobile technologies, cloud services, and geographic information systems have made it possible to develop location-based applications that simplify navigation and route planning. These technologies can be used to transform scattered transportation knowledge into a structured and accessible digital resource.

In response to this need, this project introduces **Mwasalati (مواصلاتي) – Yemeni Bus Guide**, a mobile application designed to help users navigate public bus routes in Sana'a. The application displays the user's current location, identifies nearby bus lines, and allows users to search for destinations and obtain the most suitable route, including transfer options when required.

The mobile application is developed using Flutter and Dart, with Firebase Firestore for cloud-based transportation data storage and Google Maps API for route visualization and mapping. The target system also includes a separate web-based administrative dashboard for managing transportation data, which is planned for implementation in the next academic semester. By organizing local transportation information into an easy-to-use Arabic mobile application, Mwasalati aims to reduce travel uncertainty, save time and effort, and contribute to the digital transformation of public transportation services in Yemen.

1.2 Problem Statement

Despite the widespread use of public buses in Sana'a, passengers continue to face significant difficulties when attempting to travel to unfamiliar destinations. The absence of a structured and reliable source of information about bus routes makes the process of using public transportation confusing and inefficient. The main problems identified are as follows:

- There is no dedicated mobile application or interactive map that provides accurate information about local bus routes in Sana'a.
- Users do not know whether a particular bus route serves the destination they intend to reach.
- Passengers often struggle to determine which bus line should be used for a specific journey.
- Boarding and drop-off points are not clearly identified or documented.
- Some trips require transfers between multiple buses, but users have no systematic way to identify these transfer points.
- Passengers rely heavily on asking drivers and passersby for directions, which may be inconvenient and unreliable.
- Google Maps does not provide comprehensive information about local public bus routes in Sana'a.
- The available transportation information is largely informal, unstructured, and may become outdated over time.
- Users may spend considerable time searching for suitable transportation options.
- Inefficient route choices can increase transportation costs and lead to unnecessary delays.
- Many users, particularly visitors and individuals unfamiliar with the city, experience confusion and uncertainty while navigating the transportation system.
- Some passengers avoid using public buses altogether despite their affordability because of the difficulty involved in planning trips.

These challenges demonstrate the need for a dedicated mobile application that organizes local bus route information and presents it in a clear and accessible manner. Such a system should enable users to search for destinations, identify nearby bus lines, determine boarding and drop-off points, and receive the most suitable route, including transfer options when required.

By addressing these issues, the proposed Mwasalati application aims to simplify public transportation usage in Sana'a, reduce travel time and cost, and improve the overall user experience.

1.3 Project Objectives

General Objective

The general objective of this project is to develop a mobile application that helps users in Sana'a identify nearby bus routes and determine the most suitable public transportation path to any destination within the city.

Specific Objectives

The specific objectives of the project are as follows:

1. To develop a mobile application that detects bus lines located near the user's current location.
2. To enable users to search for destinations and landmarks within Sana'a.
3. To determine the most suitable route to the selected destination, whether through a direct bus line or a journey that requires one or more transfers.
4. To display bus routes, boarding points, drop-off points, and walking directions on an interactive map.
5. To provide an Arabic-first user interface that is intuitive and accessible to local users.
6. To store and manage transportation data, including cities, bus lines, route directions, and landmarks, using Firebase Firestore.To reduce the time, effort, and transportation costs associated with navigating the city using public buses.
7. To establish a scalable digital foundation that can be extended to support additional Yemeni cities in future versions of the application.

1.4 Project Scope and Constraints

This project focuses on the development of a mobile application that supports public transportation navigation within the city of Sana'a. The scope of the system is defined as follows:

System Includes:

- Displaying the user's current location on an interactive Google Map.
- Detecting and displaying nearby bus routes based on the user's location.
- Searching for destinations and landmarks within Sana'a.

- Determining the most suitable route to the selected destination.
- Supporting both direct routes and routes that require transfers between multiple bus lines.
- Identifying boarding points and drop-off points for each trip.
- Providing walking directions from the user's location to the appropriate boarding point.
- Displaying route details and visualizing paths using map polylines.
- Providing a fully Arabic user interface with right-to-left (RTL) support.
- Allowing users to save favorite bus routes.
- Checking internet connectivity status when the application starts.
- Storing and retrieving transportation data using Firebase Firestore.
- Providing a separate web-based administrative dashboard for the Super Admin to manage transportation data, including cities, bus lines, route directions, landmarks, and transportation dataset versions.
- Supporting administrative authentication for the Super Admin to access the web-based dashboard and manage transportation data.

System Limitations (Not Included):

- Real-time tracking of bus locations.
- Fixed schedules for bus arrival and departure times.
- Electronic payment functionality.
- Automatic fare calculation.
- Support for iOS devices.
- Coverage of cities other than Sana'a.
- Advanced artificial intelligence features.

Constraints:

- The accuracy of route recommendations depends on the quality and completeness of the transportation data collected manually.
- Bus routes may change over time, requiring regular updates to the database.
- The application depends on third-party services such as Google Maps API and Firebase Firestore.
- Access to cloud-based data generally requires an internet connection.
- Internet connectivity in Yemen may be unstable or limited in some areas.
- The current version is designed primarily for Android devices only.

1.5 Project Significance

This project is significant because it provides a practical and innovative solution to a common transportation problem faced by public bus users in Sana'a. By offering a digital platform that organizes bus route information and presents it through an easy-to-use mobile application, Mwasalati helps users plan their trips more efficiently and confidently.

The proposed system supports passengers in reaching their destinations with less effort by reducing the uncertainty associated with identifying suitable bus routes. It minimizes the time spent searching for transportation options, decreases the likelihood of taking inefficient routes, and helps lower transportation costs. These improvements enhance the overall travel experience and encourage greater use of public transportation.

From a societal perspective, the project contributes to the digital transformation of local transportation services in Yemen. It converts informal route knowledge, which is traditionally communicated verbally, into a structured and searchable digital database. This organized information can serve as a foundation for future smart transportation initiatives and support broader efforts to modernize public services.

From an academic perspective, the project demonstrates the practical application of several Information Technology concepts, including mobile application development, database design, geographic information systems (GIS), software engineering, and route planning algorithms. It illustrates how theoretical knowledge can be applied to address a real-world transportation problem in the local community.

The project uses modern technologies such as Flutter, Firebase Firestore, and Google Maps API to establish a structured and maintainable foundation for public bus transportation guidance. The target system architecture also provides for a separate web-based administrative dashboard and a versioned transportation dataset, supporting future maintenance and potential expansion to other Yemeni cities.

Key Benefits:

- Improved accessibility to public transportation information in Sana’a.
- Reduced travel time, effort, and transportation costs for users.
- Increased confidence and convenience when using public buses.
- Support for the digital transformation of transportation services in Yemen.
- Practical application of modern mobile and cloud technologies.
- Scalable foundation for expansion to other Yemeni cities.
- Potential for future enhancements such as real-time tracking and fare estimation.

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1.6 Project Feasibility

A feasibility study was conducted to evaluate whether the Mwasalati application can be successfully developed, deployed, and used as a practical solution for public transportation guidance in Sana’a. The purpose of this analysis is to assess the viability of the project from economic, technical, and operational perspectives. The study confirms that the proposed system is realistic and achievable using currently available technologies and resources, while also addressing a genuine transportation need within the local community.

1.6.1 Economic Feasibility

The Mwasalati project is economically feasible because it is developed using modern technologies that are available at no cost or offer generous free usage tiers. Flutter and Dart are open-source development tools, while Firebase Firestore and Google Maps API provide usage-based pricing and, where applicable, free usage quotas that may help reduce the initial development and deployment costs. Actual costs depend on service usage, applicable pricing plans, and the requirements of the deployed system.

The system does not require dedicated on-premises servers or specialized hardware infrastructure. Users only need an Android smartphone and internet connectivity to access the application. From the developer’s perspective, the application can be deployed and maintained with relatively low operational costs during the early stages of adoption.

If the application attracts a large number of active users, additional costs may arise due to increased usage of third-party services such as Google Maps API and Firebase. However, these costs are usage-based and would only become relevant once the application demonstrates practical value and user demand. In addition to its low implementation cost, the system offers economic benefits to end users by reducing unnecessary travel time and transportation expenses resulting from inefficient route selection.

1.6.2 Technical Feasibility

The project is technically feasible because it is implemented using well-established and widely supported technologies. The mobile application is developed with Flutter and Dart, Firebase Firestore is used for cloud-based data management, and Google Maps API provides interactive mapping and geographic visualization capabilities. These technologies are mature, reliable, and extensively documented, making them suitable for the development of scalable mobile applications.

The project requirements are fully achievable within the capabilities of the selected technology stack. Route planning is implemented using polyline analysis and custom search algorithms that identify direct and transfer-based journeys. These algorithms are computationally practical and compatible with mobile devices.

In addition, the development team possesses the technical knowledge required to design, implement, and integrate the system components. The use of structured software engineering practices and modular architecture further supports maintainability and future enhancement.

1.6.3 Operational Feasibility

The Mwasalati application is operationally feasible because it addresses a real and frequently encountered transportation problem in Sana’a. Users often struggle to determine which bus routes can take them to their destinations, and the application provides a clear and convenient solution to this issue.

The system is designed with an Arabic-first user interface that aligns with the language preferences of the target audience. The application is intended to be intuitive and straightforward, allowing users to interact with it without the need for formal training or technical expertise.

The target system also includes a web-based administrative dashboard intended for use by the authorized Super Admin. The dashboard is planned for implementation in the next academic semester and is expected to support the structured management of transportation data.

The widespread use of Android smartphones among students, employees, and other members of the community increases the practicality of deploying the application. Since the system offers immediate and tangible benefits, it is expected to achieve a high level of user acceptance and adoption.

Overall, the feasibility analysis demonstrates that Mwasalati is economically affordable, technically achievable, and operationally practical. These findings confirm that the project represents a viable solution for improving access to public transportation information in Sana'a and provides a solid foundation for future expansion and enhancement.

1.7 Project Schedule

1.7.1 Overview

The development of the Mwasalati application was planned using an Agile Software Development approach, which emphasizes incremental delivery, continuous testing, and iterative refinement. This methodology was selected because it allows the system to be developed in manageable stages, with each phase producing a functional component that can be evaluated and improved before proceeding to the next stage.

For academic planning purposes, the project schedule is organized over a twelve-week period. This timeline represents a realistic and structured development plan for the analysis, design,

implementation, testing, and documentation of the system. Each phase includes specific objectives and deliverables that contribute to the successful completion of the project.

1.7.2 Phase 1 – Project Initiation and Requirements (Week 1–2)

This phase focused on designing the overall system architecture and data structure. The Firestore database collections were defined, and the main user interfaces were planned. Initial wireframes and design decisions were prepared to guide the implementation process.

Deliverables:

- System architecture design
- Firestore data model
- UML diagrams
- UI wireframes

1.7.3 Phase 2 – System Analysis and Design (Week 3–4)

In this phase, the Flutter project was initialized and integrated with Firebase Firestore and Google Maps API. Core services, including location detection, connectivity monitoring, and map rendering, were developed.

Deliverables:

- Project setup
- Firebase integration
- Google Maps integration
- Core services implementation

1.7.4 Phase 3 – Core Implementation (Week 5–6)

This phase involved implementing the application's main functional logic. Algorithms were developed to detect nearby bus lines, search destinations, and calculate both direct and transfer-based routes.

Deliverables:

- Nearby bus detection
- Destination search
- Direct route planning
- Transfer route planning

1.7.5 Phase 4 – Route Planning and Search Implementation (Week 7–8)

This phase focuses on completing the implementation of the route planning and search functionalities. It includes integrating nearby bus line detection, destination search, direct route identification, and transfer-based route planning with the application's map interface. The implemented functionality is prepared for subsequent testing and refinement.

Deliverables:

- Nearby bus line detection integration
- Destination search and landmark selection
- Direct and transfer-based route planning integration
- Route visualization and trip details

1.7.6 Phase 5 – Testing and Refinement (Week 9–10)

This phase focuses on testing the implemented mobile application functionalities and refining the system based on the identified issues. Functional testing is conducted to verify the behavior of the main features, while detected errors are corrected and the user interface is refined to improve usability and consistency.

Deliverables:

- Functional testing activities and documented results
- Identified and corrected software defects
- User interface refinements
- Testing and refinement documentation

1.7.7 Phase 6 – Documentation and Final Delivery (Week 11–12)

The final phase focused on preparing the project report, organizing screenshots and diagrams, and developing the presentation materials required for project submission and defense.

Deliverables:

- Final project report
- Presentation slides
- System screenshots
- Final project package

1.7.8 Schedule Summary

Weeks	Phase	Main Outputs
Week 1–2	Initiation and Requirements	Problem definition, requirements, scope
Week 3–4	System Analysis and Design	Architecture, database design, UML diagrams
Week 5–6	Core Implementation	Flutter setup, Firebase and Maps integration
Week 7–8	Route Planning and Search	Integrated destination search, nearby bus line detection, direct and transfer-based route planning, and route visualization.
Week 9–10	Testing and Refinement	Functional testing, documented results, defect corrections, and user interface refinements.
Week 11–12	Documentation and Delivery	Final report, presentation, submission materials

The proposed schedule demonstrates that the project can be completed within a reasonable academic timeframe using an iterative development methodology. By dividing the work into clearly defined phases, the development process remains organized, manageable, and aligned with the objectives of the project.

1.8 Project Requirements

The successful development and operation of the Mwasalati application require a combination of appropriate hardware resources, software tools, cloud services, and development technologies. These components provide the technical foundation needed to build, test, deploy, and operate the application effectively.

1.8.1 Hardware Requirements

The following hardware resources were used during the development of the project:

1. A laptop or desktop computer for coding, testing, and debugging the application.

2. An Android smartphone for running and testing the application on a physical device.
3. A stable internet connection to access cloud services, APIs, and development resources.

The following hardware and connectivity requirements are necessary for end users to operate the application:

1. An Android smartphone.
2. A GPS-enabled mobile device to determine the user's current location.
3. An internet connection to retrieve map data and access cloud-based transportation information.

For the administrative dashboard, an internet-connected computer with a modern web browser is required for the authorized Super Admin to access and manage transportation data through the web interface.

1.8.2 Software Requirements

The following software tools were used during development:

1. Flutter SDK for cross-platform mobile application development.
2. Dart SDK as the programming language environment.
3. Android Studio for Android development, emulation, and debugging.
4. Visual Studio Code as the primary source code editor.
5. Git and GitHub for version control and source code management.

The project relies on several external services and APIs:

1. Firebase Core for initializing Firebase services.
2. Firebase Firestore for cloud-based NoSQL data storage.
3. Google Maps API for map rendering and geographic visualization.

Several Flutter packages were used to implement core functionalities:

1. google_maps_flutter for embedding interactive maps.
2. geolocator for obtaining device location and calculating distances.
3. connectivity_plus for monitoring network connectivity status.

The project supports the following operating systems:

1. Windows for development.
2. Android for application deployment and usage.

1.8.3 Technologies Used

- Flutter
- Dart
- Firebase Firestore
- Firebase Core
- Google Maps API
- google_maps_flutter
- geolocator
- connectivity_plus
- Clean Architecture
- Repository Pattern
- SOLID Principles
- Agile Modeling (AM)
- Scrumban
- Unified Modeling Language (UML)
- Entity-Relationship Diagram (ERD)

The selected technology stack provides a modern, scalable, and maintainable foundation for the development of the Mwasalati application. These technologies were chosen for their reliability, extensive documentation, and suitability for building location-based mobile applications that address real-world transportation challenges.

1.9 Project Organization

The Mwasalati project was developed as an individual graduation project by the student under the academic supervision of a faculty member at the Lebanese International University (LIU) – Yemen. The project organization was structured to ensure that all activities, from initial problem identification to final documentation, were carried out in a systematic and coordinated manner.

As the sole developer of the project, the student was responsible for all major phases of the system development life cycle. These responsibilities included identifying the transportation problem, gathering and analyzing requirements, reviewing related literature, designing the

system architecture and database structure, implementing the mobile application, conducting system testing, and preparing the final report and presentation materials.

The project was managed using an Agile development approach, which allowed the work to be completed incrementally through a series of iterative stages. This approach supported continuous refinement of the application and enabled improvements to be incorporated throughout the development process. To organize the source code and track development progress, GitHub was used for version control, while local files were maintained to document ideas, notes, and task planning.

The project was conducted under the guidance of the academic supervisor, who provided feedback and recommendations during different stages of development. This supervision helped ensure that the project remained aligned with academic standards and fulfilled the requirements of the Bachelor of Science in Information Technology program.

1.9.1 Student Responsibilities

The student was responsible for the following activities:

- Problem identification and requirements analysis.
- Literature review and benchmarking of similar systems.
- System design, including architecture, data modeling, and UML diagrams.
- Mobile application development using Flutter and Firebase.
- Integration of Google Maps and location services.
- Functional testing, debugging, and performance refinement.
- Preparation of the final project report and presentation.

1.9.2 Development Workflow and Coordination

Although the project was completed individually, development was organized using a disciplined workflow based on iterative planning and regular progress reviews. Each development cycle focused on a specific module, such as route search, map integration, or user interface design. Feedback from testing and supervisor reviews was used to refine the system and improve its quality.

This structured approach helped maintain consistency across all project components and reduced the risks associated with developing a complex location-based mobile application.

1.9.3 Report Organization

The remainder of this report is organized as follows:

- **Chapter 1: Introduction** – Presents the project background, problem statement, objectives, scope, significance, feasibility, schedule, and requirements.
- **Chapter 2: Literature Review** – Reviews existing transportation and navigation systems and compares related applications.
- **Chapter 3: Methodology and System Analysis** – Describes the development methodology, requirements analysis, and system design models.
- **Chapter 4: Implementation and Testing** – Explains the implementation details, technologies used, user interfaces, and testing results.
- **Chapter 5: Conclusion and Future Work** – Summarizes the project outcomes, limitations, and recommendations for future enhancements.

The project organization adopted for Mwasalati ensured effective planning, systematic development, and comprehensive documentation, contributing to the successful delivery of a practical solution for public transportation navigation in Sana'a.

Chapter 2
Review of Literature

2.1 Introduction

This chapter presents a review and comparative analysis of three widely used transportation applications: Google Maps, Moovit, and Citymapper. These systems were selected because they represent mature and well-established solutions for public transportation guidance and route planning. Each application provides users with tools to search for destinations, identify available bus routes, and navigate through urban transportation networks. Examining these systems provides valuable insight into the functional and technical approaches used to deliver location-based transportation services to a large and diverse user base.

The comparison focuses on key aspects such as core functionality, geographic coverage, and support for bus transportation. Particular attention is given to the ability of these systems to provide route recommendations, transfer planning, and map-based trip visualization. Although these applications offer powerful and highly refined features, they do not provide structured and reliable coverage of public bus routes in Sana'a, where transportation information remains largely informal and undocumented. This gap highlights the need for a dedicated solution tailored to the Yemeni context. The findings of this review have helped identify the most relevant design practices and functional requirements adopted in the development of Mwasalati, which aims to provide an Arabic-first mobile application for public bus guidance in Yemen.

2.2 Similar Projects (Benchmarking)

2.2.1 System 1: Google Maps

Google Maps is one of the most widely used mapping and navigation applications in the world. Developed by Google, the application provides comprehensive geographic information, turn-by-turn navigation, and multimodal route planning for driving, walking, cycling, and public transportation. In many countries, Google Maps includes detailed transit information such as bus routes, stop locations, and estimated travel times. Its global reach and extensive feature set make it an important benchmark for transportation-related applications. Google Maps is also available in Arabic and is widely used by smartphone users in Yemen for general navigation purposes.

Objectives

The primary objective of Google Maps is to help users navigate efficiently between locations by providing accurate mapping data, real-time traffic information, and optimized route recommendations. In the context of public transportation, the application aims to simplify trip

planning by identifying available transit options and presenting alternative routes based on distance, estimated duration, and transfer requirements.

Technologies Used

- Mobile Application: Native Android and iOS applications.
- Web Application: JavaScript-based interactive mapping platform.
- Backend Infrastructure: Google Cloud distributed services.
- Data Sources: Satellite imagery, GPS data, road networks, and user-contributed updates.
- Geographic Technologies: Geographic Information Systems (GIS) and digital cartography.
- Artificial Intelligence: Machine learning algorithms for traffic prediction and route optimization.

Main Features

- Search for addresses, landmarks, and points of interest.
- Interactive map visualization.
- Public transportation route planning.
- Walking directions to and from transit stops.
- Alternative route suggestions.
- Estimated travel times.
- Real-time traffic updates.
- Multilingual support, including Arabic.
- Saved places and favorites.

Advantages

- Provides highly accurate and continuously updated geographic data.
- Offers an intuitive and well-designed user interface.
- Supports multiple transportation modes within a single application.
- Includes Arabic language support.
- Integrates destination search, route planning, and map visualization.
- Widely available and trusted by millions of users worldwide.

Disadvantages

- Does not provide comprehensive and structured bus route information for Sana'a.
- Relies on official transit datasets that are unavailable for most Yemeni cities.
- Does not document informal boarding and drop-off locations used by local buses.
- Limited customization for the specific characteristics of Yemen's transportation system.
- Requires an internet connection to access most route-planning features.

Conclusion

Google Maps provides a powerful and reliable navigation platform that combines destination search, interactive maps, and multimodal route planning. However, its public transportation capabilities depend on the availability of structured transit data, which is currently lacking in Yemen. As a result, the application does not adequately support users who need detailed information about local bus routes in Sana'a. This limitation underscores the need for a specialized solution such as Mwasalati, which is specifically designed to organize and present public transportation information tailored to the Yemeni environment.

2.2.2 System 2: Moovit

Moovit is a specialized public transportation application designed to help users navigate urban transit systems using buses, metro, trains, and other mobility options. The platform is widely recognized as one of the leading transit applications in the world and provides service in thousands of cities across more than one hundred countries. Unlike general navigation applications, Moovit focuses specifically on public transportation and combines official transit data with community-generated updates to improve route accuracy and service awareness. The application is available in Arabic and is widely used in the Middle East and other regions.

Objectives

The primary objective of Moovit is to simplify the use of public transportation by providing users with accurate route planning, real-time arrival information, and live service alerts. The system aims to reduce uncertainty during travel and help passengers select the most efficient route based on current transit conditions and available transportation alternatives.

Technologies Used

- Mobile Application: Native Android and iOS applications.
- Frontend Technologies: Modern mobile user interface frameworks.
- Backend Infrastructure: Cloud-based distributed services.
- Data Sources: General Transit Feed Specification (GTFS), GPS data, and crowdsourced updates.
- Geographic Technologies: GIS and map-based visualization services.
- Analytics: Data processing and predictive algorithms for arrival estimation.

Main Features

- Public transportation route planning.
- Real-time arrival predictions in supported cities.
- Live service alerts and disruption notifications.
- Nearby stops and station detection.
- Step-by-step navigation guidance.
- Multimodal trip planning.
- Community-based updates and corrections.
- Arabic language support.
- Favorites and frequently used locations.

Advantages

- Designed specifically for public transportation rather than general navigation.
- Provides detailed transit guidance with transfer planning.
- Supports Arabic and many other languages.
- Combines official transit data with crowdsourced information.
- Offers real-time arrival estimates and service alerts in supported cities.
- Covers a very large number of cities worldwide.

Disadvantages

- Data quality depends on the availability and accuracy of local transit information.
- Some advanced features are unavailable in cities without real-time data feeds.

- Does not provide dedicated and verified coverage for bus routes in Sana'a.
- Free versions may include advertisements.
- Requires an internet connection for most route-planning and live-update features.

Conclusion

Moovit represents a highly specialized and mature public transportation platform that offers advanced trip planning and transit-focused navigation features. Its integration of official transit data, crowdsourced updates, and multilingual support demonstrates effective approaches for organizing transportation information at scale. However, the application does not currently provide comprehensive and validated coverage of public bus routes in Sana'a. This limitation highlights the need for a localized solution such as Mwasalati, which is specifically designed to structure and present transportation information tailored to the Yemeni context.

2.2.3 System 3: Citymapper

Citymapper is a public transportation application designed to provide detailed route planning for major metropolitan areas. Developed by Citymapper, the application focuses on helping users navigate complex urban transportation networks by integrating buses, metro systems, trains, walking, cycling, and ride-hailing services. Citymapper is widely recognized for its clean user interface, accurate transit guidance, and highly detailed step-by-step instructions. Although its geographic coverage is more limited than that of Google Maps and Moovit, it offers a sophisticated transit experience in the cities it supports.

Objectives

The primary objective of Citymapper is to simplify urban mobility by providing intelligent and user-friendly route recommendations across multiple transportation modes. The system aims to help users identify the fastest, most convenient, and most cost-effective way to reach their destinations while minimizing uncertainty during travel.

Technologies Used

- Mobile Application: Native Android and iOS applications.
- Web Application: Browser-based trip planning platform.

- Backend Infrastructure: Cloud-based distributed services.
- Data Sources: GTFS transit feeds, GPS data, and real-time service information.
- Geographic Technologies: GIS and interactive map-rendering technologies.
- Optimization Algorithms: Routing and journey-ranking algorithms for multimodal transportation.

Main Features

- Public transportation route planning.
- Step-by-step navigation instructions.
- Real-time departure and arrival information.
- Transfer planning across multiple transit modes.
- Estimated trip duration and cost.
- Nearby stop and station detection.
- Service disruption alerts.
- Favorites and saved destinations.

Advantages

- Provides highly detailed and easy-to-follow trip instructions.
- Offers a modern and intuitive user interface.
- Includes sophisticated transfer and multimodal planning capabilities.
- Presents estimated trip duration and cost information.
- Delivers excellent user experience in supported cities.

Disadvantages

- Geographic coverage is limited to selected major cities.
- Does not provide service coverage for Sana’a or other Yemeni cities.
- Arabic language support is limited compared with Google Maps and Moovit.
- Depends heavily on structured transit datasets and real-time feeds.
- Requires an internet connection for route planning and live updates.

Conclusion

Citymapper offers an advanced and user-centered approach to public transportation guidance, particularly in large metropolitan areas with well-structured transit data. Its strong emphasis on usability and detailed trip instructions provides valuable design insights for transportation applications. However, the system is unavailable in Yemen and depends on transit infrastructure and datasets that do not currently exist in Sana'a. These limitations highlight the importance of developing a localized application such as Mwasalati, which is specifically designed to address the characteristics and information needs of Yemen's public bus network.

2.3 Comparison Table

Table 0.1 Comparison Table of the Three Systems

Criteria	**Google Maps**	**Moovit**	**Citymapper**	**Mwasalati**
Platform	Android, iOS, Web	Android, iOS, Web	Android, iOS, Web	Android
Geographic Coverage	Global	Global (selected transit-enabled cities)	Major metropolitan cities	Sana'a
Nearby Route Detection	Yes	Yes	Yes	Yes
Destination Search	Yes	Yes	Yes	Yes
Transfer Planning	Yes	Yes	Yes	Yes
Interactive Map	Yes	Yes	Yes	Yes
Arabic Language Support	Yes	Yes	Limited	Yes

Criteria	Google Maps	Moovit	Citymapper	Mwasalati
Offline Support	Limited	Limited	Limited	Limited (local caching planned)
Real-Time Tracking	Yes (where data is available)	Yes (supported cities)	Yes (supported cities)	No
Fare Estimation	Limited	Limited	Yes	No
AI Features	Yes	Limited	Limited	Planned
Yemen-Specific Coverage	No	No	No	Yes
Localized for Informal Bus Networks	No	No	No	Yes

2.4 Conclusion

The review of Google Maps, Moovit, and Citymapper highlights a range of advanced approaches to public transportation guidance and urban navigation. These systems demonstrate the effectiveness of integrating interactive maps, destination search, route planning, and transfer-based journey recommendations within user-friendly mobile applications. In addition, they illustrate how real-time data and large-scale geographic information systems can significantly improve the overall transportation experience in cities where structured transit datasets are available.

Despite their technical sophistication, the reviewed systems do not adequately address the needs of public bus users in Sana'a. Their transit functionality depends on the availability of organized transportation data, which is largely unavailable in Yemen, and they are not specifically

designed to support informal bus networks that operate without fixed schedules. Mwasalati addresses this gap by organizing locally collected transportation data into a dedicated mobile application that identifies nearby bus routes, supports destination search, and determines both direct and transfer-based journeys. By focusing on the operational realities of public transportation in Yemen, the system provides a practical and scalable foundation for extending digital bus guidance services to other Yemeni cities in future versions